



Module 8 Day 2 : Ports

Activity File: Ports

In this exercise, you'll continue to play the role of a security analyst at Acme Corp.

- You will analyze several network requests captured from the potentially rogue employee, Sally Stealer.
- Your manager tasks you with finding the source and destination port for each request and determining the protocol for each destination port. You will use your findings to determine what kind of activities Sally Stealer may be conducting.

Instructions

1. View the log records listed below, and note that each log record is distinguished with a title: Log Record 1, Log Record 2, etc.
2. Determine the source and destination port for each request.
3. Determine the protocol associated with each destination port.
4. Research each protocol and summarize what Sally Stealer might be doing based on your findings.

Log Record 1:

=====

Src: 192.168.1.9, Dst: 192.124.249.168

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 942

Identification: 0x4103 (16643)

Flags: 0x4000, Don't fragment

Time to live: 128

Protocol: TCP (6)

Header checksum: 0x3a70 [validation disabled]

[Header checksum status: Unverified]

Source: 192.168.1.9

Destination: 192.124.249.168

Src Port: 50152, Dst Port: 80, Seq: 3879, Ack: 39771, Len: 902

Source Port: 50152

Destination Port: 80

[Stream index: 82]

[TCP Segment Len: 902]

Sequence number: 3879 (relative sequence number)

[Next sequence number: 4781 (relative sequence number)]

Acknowledgment number: 39771 (relative ack number)

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

Window size value: 256

[Calculated window size: 65536]

Log Record 2

=====

Src: 10.0.0.42, Dst: 35.186.241.40

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 86

Identification: 0x0211 (529)

Flags: 0x4000, Don't fragment

Time to live: 128

Protocol: TCP (6)

Header checksum: 0xd984 [validation disabled]

[Header checksum status: Unverified]

Source: 10.0.0.42

Destination: 35.186.241.40

Src Port: 53367, Dst Port: 443, Seq: 1136, Ack: 1307, Len: 46

Source Port: 53367

Destination Port: 443

[Stream index: 113]

[TCP Segment Len: 46]

Sequence number: 1136 (relative sequence number)

[Next sequence number: 1182 (relative sequence number)]

Acknowledgment number: 1307 (relative ack number)

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

Window size value: 251

[Calculated window size: 64256]

[Window size scaling factor: 256]

Checksum: 0xc5d7 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

[SEQ/ACK analysis]

[Timestamps]

TCP payload (46 bytes)

Log Record 3

=====

Src: 192.168.29.54, Dst: 90.130.70.73

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x10 (DSCP: Unknown, ECN: Not-ECT)

Total Length: 59

Identification: 0xcccd (52429)

Flags: 0x4000, Don't fragment

Time to live: 64

Protocol: TCP (6)

Header checksum: 0xef35 [validation disabled]

[Header checksum status: Unverified]

Source: 192.168.29.54

Destination: 90.130.70.73

Src Port: 64836, Dst Port: 21, Seq: 17, Ack: 55, Len: 7

Source Port: 64836

Destination Port: 21

[Stream index: 2]

[TCP Segment Len: 7]

Sequence number: 17 (relative sequence number)

[Next sequence number: 24 (relative sequence number)]

Acknowledgment number: 55 (relative ack number)

1000 = Header Length: 32 bytes (8)

Flags: 0x018 (PSH, ACK)

000. = Reserved: Not set

...0 = Nonce: Not set

.... 0... = Congestion Window Reduced (CWR): Not set

.... .0.. = ECN-Echo: Not set

.... ..0. = Urgent: Not set

.... ...1 = Acknowledgment: Set

.... 1... = Push: Set

....0.. = Reset: Not set

....0. = Syn: Not set

....0 = Fin: Not set

[TCP Flags:AP...]

Window size value: 229

[Calculated window size: 29312]

[Window size scaling factor: 128]

Checksum: 0xcb58 [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

[SEQ/ACK analysis]

[Timestamps]

TCP payload (7 bytes)